Candidate Data Transformation Pipeline

Technical Design Overview | Rikesh Yadav | rikeshyadav2780@gmail.com

OBJECTIVE Design a deterministic, explainable pipeline that transforms heterogeneous candidate data from multiple sources into a unified canonical profile. The system normalises data, resolves conflicts via confidence-based merge strategies, preserves field-level provenance, and validates output.

HIGH-LEVEL ARCHITECTURE

Recruiter CSV | ATS JSON | GitHub Profile

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Input Detection & Routing

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Data Extraction

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Canonical Candidate Model

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Data Normalisation (phone / date / skill)

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Entity Resolution & Merge Engine
- Confidence Scoring
- Field-Level Provenance

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Projection / Mapping Layer

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Schema Validation Layer

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Standardised JSON Output

CORE DESIGN PRINCIPLES

1 Canonical Data Model

- Single internal representation independent of source format
- Decouples source parsing from all downstream business logic
- New source type = one new extractor, zero other changes
- Enables independent testing of each pipeline stage

2 Deterministic Merge Strategy

- Trust-weighted conflict resolution — highest trust source wins
- Same inputs always produce exactly the same output
- Cross-source agreement detection boosts field confidence
- Conflict penalty preserves honest signal over inflated scores

3 Field-Level Provenance

- Every field stores: source, method, and a plain-English reason
- Reason explains WHY that value and source were chosen
- Alternatives surfaced in provenance, never silently discarded
- Complete audit trail visible in the output JSON itself

4 Projection / Mapping Layer

- Canonical model is NEVER modified by output shaping
- Config: rename fields, subset, normalise, on_missing
- Same engine — multiple consumer views from one config

5 Validation Layer

- Required fields, schema types, and formats enforced
- Invalid records surface warnings — never silent bad output

FIELD CONFIDENCE SCORE

Field confidence = min(1.0,

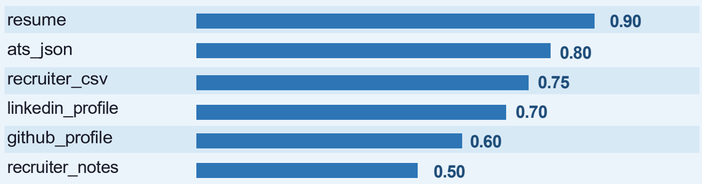
source_trust resume 0.90 | ats 0.80 | csv 0.75 | gh 0.60

+ agreement_bonus +0.03 per confirming independent source

+ validation_bonus +0.02 if value passes format check

- conflict_penalty -0.10 if another source disagreed)

SOURCE TRUST MATRIX



EXTENSIBILITY & FUTURE SCOPE

- Plug-in extractor interface (LinkedIn, Resume PDF/DOCX)
- Configurable trust weights per org / deployment
- Additional output formats (CSV, Avro, Parquet)
- REST API or Web UI integration layer
- AI-assisted field enrichment and entity resolution
- Batch mode for thousands of candidates